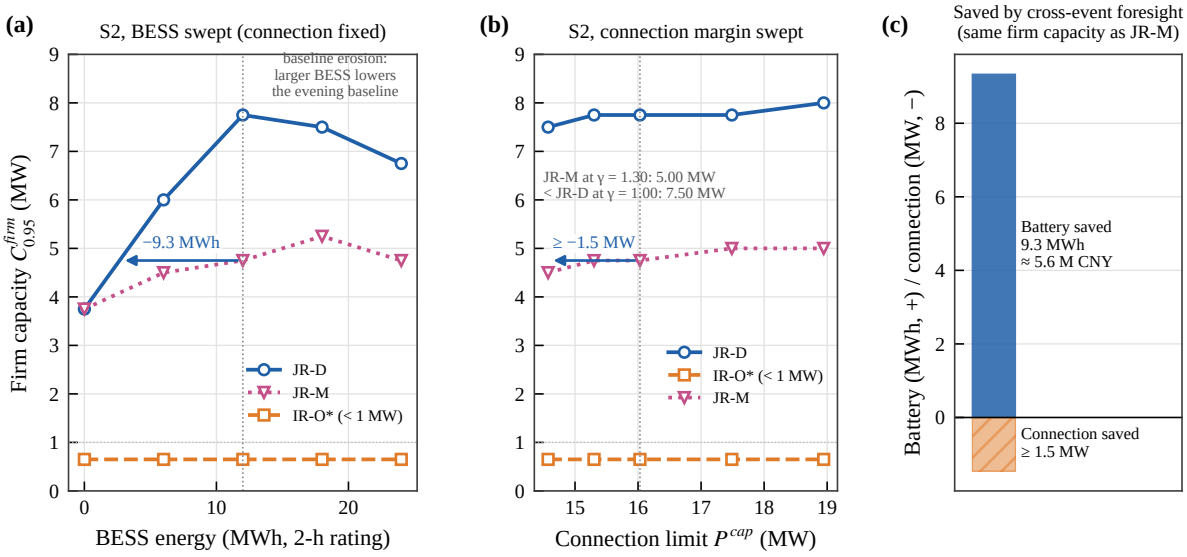
(a)
S2, BESS swept (connection fixed)
baseline erosion:
larger BESS lowers
the evening baseline
−9.3 MWh
JR-D
JR-M
IR-O* (< 1 MW)
Firm capacity $C_{0.95}^{firm}$ (MW)
BESS energy (MWh, 2-h rating)
(b)
S2, connection margin swept
JR-M at γ = 1.30: 5.00 MW
< JR-D at γ = 1.00: 7.50 MW
≥ −1.5 MW
JR-D
IR-O* (< 1 MW)
JR-M
Connection limit P^{cap} (MW)
(c)
Saved by cross-event foresight
(same firm capacity as JR-M)
Battery saved
9.3 MWh
≈ 5.6 M CNY
Connection saved
≥ 1.5 MW
Battery (MWh, +) / connection (MW, −)